

Phil Reny route: summary for a math major

This note explains the earlier proof route based on Phil Reny’s suggestion. It is meant to be read separately from the final menu-engine summary. The point is not to record the sequence of runs, but to explain the idea, why it was promising, what it achieved conditionally, and where it stalled.

1. The motivation

The original finite proof of Theorem 2 uses a zero-sum game between the agent and the misaligned adviser. In finite spaces, the agent and adviser strategy sets are products of compact simplices. Sion’s minimax theorem gives a saddle point, and that saddle point gives robust rationalizability.

In infinite M , this direct route breaks. The adviser can choose arbitrary measurable kernels $\beta : M \rightarrow \Delta(M)$, and the agent can choose measurable kernels $\sigma : M \times \Theta \rightarrow \Delta(A)$. These spaces are too large for the finite proof to carry over directly. Continuity in the adviser kernel is especially problematic because messages affect the agent’s continuation payoff endogenously.

Phil Reny’s suggestion was to split the problem into two stages:

1. First prove the existence of an optimal agent strategy by restricting the adversary to a more regular class of kernels.
2. Then try to lift the result back to the full class of measurable adversaries.

This was a serious route because it targeted the main compactness problem directly, rather than trying to force the whole infinite adviser space to be compact.

2. Stage 1: restrict the adversary

The first move was to restrict the misaligned adviser to kernels that are absolutely continuous with respect to τ :

$$\beta_\varphi(dm \mid s) = \varphi(m \mid s)\tau(dm).$$

Here φ is a jointly measurable density, normalized in m for each source belief s .

Why does this help?

Because after this restriction, the reported-message variable m is integrated against a fixed base measure $\tau(dm)$. This fixed marginal is what makes the agent side manageable. One can use Balder’s weak topology for transition probabilities: roughly, a sequence of agent kernels converges if the induced joint laws over messages, private types, and actions converge in the right stable sense.

The important technical facts are:

- the agent strategy space is compact in this Balder topology, after the usual identification of strategies that agree almost everywhere;
- for each fixed restricted adviser density φ , the payoff is continuous in the agent strategy;
- the restricted adviser class is convex.

Mertens’s asymmetric minimax theorem then gives a maximizer σ^* for the restricted game:

$$\max_{\sigma} \inf_{\varphi} U_F(\sigma, \varphi).$$

At this point, we have a strong candidate agent strategy, but only for the restricted game where the adviser must use τ -dominated kernels.

3. Stage 2: Lusin lift to unrestricted adversaries

The next step was to show that the same σ^* also works against arbitrary measurable adviser kernels β .

The idea uses Lusin’s theorem. A measurable function need not be continuous everywhere, but it is continuous on large compact sets. The agent’s strategy, as a function of the message, can be represented as continuous on an increasing sequence of compact sets

$$K_1 \subseteq K_2 \subseteq \dots, \quad K^* = \bigcup_n K_n,$$

with full measure.

If these compact sets are also **thick** for all the state-conditional laws, then an arbitrary adversary message can be slightly “smeared” into a nearby τ -dominated distribution without changing the payoff much. In intuitive terms: if the adviser wants to send a message y , replace that point message by a small cloud of nearby messages inside the compact set. Continuity of the payoff on the compact set makes this cloud almost as good as the original message.

This is the Lusin-lift step. It says that every unrestricted adversary can be approximated, at σ^* , by a restricted τ -dominated adversary. Therefore the restricted-game maximizer also secures the full robust value.

4. The first added condition: Lusin thickness

The Lusin lift needs more than ordinary full measure. It needs the compact sets K_n to have enough local mass for every state-conditional signal law. Otherwise the “small cloud around a message” might have zero mass under some relevant state.

This led to the condition called **A5-thick**.

In simple terms, A5-thick says that the Lusin compact sets are not just large measure-theoretically; they are locally visible to all the state-conditional posterior laws.

Why is some condition like this needed? Consider a perfect-revelation example with two states. In state 0, the adviser belief is always δ_0 ; in state 1, it is always δ_1 . The two state-conditional laws live on disjoint points. There is no common locally thick support on which one can smooth messages in a way visible from every state. The Lusin smoothing mechanism fails.

So the route proved value optimality only under this extra thickness condition.

5. The second added condition: exact rowwise attainment

Even after finding a value-optimal agent strategy, Theorem 2 needs an actual worst-case adviser β^* , not merely approximations.

For a fixed source belief s , the adviser wants to choose a message m that minimizes the agent’s payoff. In formulas, one studies a row function

$$\ell(m, s),$$

the payoff from sending message m when the source belief is s .

The problem is that the infimum over m may not be attained. For example, a function can have infimum 0 on an interval without ever taking the value 0. Then the adversary can get arbitrarily close to the worst case, but no exact minimizing message exists.

This led to **A8c-attain**: for almost every s , the rowwise infimum is attained at some message, and these minimizing messages can be selected measurably.

Under A8c-attain, one can define

$$\beta^*(\cdot \mid s) = \delta_{m^*(s)}$$

and obtain an exact adversary.

6. The third added condition: posterior calibration

Even with an exact worst-case adviser, robust rationalizability still needs one more property. After a message m , the agent forms a posterior using Bayes' rule and the conjectured adviser strategy β^* . The prescribed continuation at m must be optimal for that posterior.

The Phil Reny route produced worst messages source by source. But it did not automatically ensure that, after grouping all sources that send the same message m , their conditional average belief lands in the right Bayes optimality cone for the action prescribed at m .

This is a mass-balance problem. The name used in that version was **TRE-gen-Hall**.

The analogy with Hall's marriage theorem is useful. We need to match source beliefs to messages so that:

- each source belief is sent to a worst-case message;
- each message receives a mixture of sources whose average belief justifies the agent's continuation at that message.

In one-dimensional or highly symmetric cases, this kind of matching can be done by monotone or radial transport. In three or more states, it becomes a vector balancing problem. The Phil Reny route did not derive this matching from the standing primitives; it had to assume it.

7. What the Phil Reny route achieved

The route achieved a conditional theorem.

Under standing assumptions plus A5-thick, the Lusin-lift argument gives a value-optimal agent strategy.

Adding A8c-attain gives an exact worst-case adviser.

Adding TRE-gen-Hall gives full robust rationalizability.

So the route did not fail because the tools were wrong. It identified a real path through the infinite-dimensional strategy problem. But it required three substantive extra assumptions:

- A5-thick, to make the Lusin smoothing work;
- A8c-attain, to turn approximate worst-case messages into exact worst-case messages;
- TRE-gen-Hall, to make the induced posteriors justify the agent's continuation strategies.

8. Why the route stalled

The route stalled because these extra assumptions were not just cosmetic.

First, A5-thick is tied to a representative of an abstract measurable strategy. It is a condition on how the chosen maximizer behaves on Lusin compact sets. That makes it less primitive than one would like.

Second, A8c-attain is a genuine attainment assumption. Without regularity, the adversary’s infimum may be approached but not attained.

Third, TRE-gen-Hall is close to the rationalizability conclusion itself. It is the condition that the worst-case routing also produces Bayes-consistent posteriors at messages. This is exactly the feature that finite Sion gives for free but the infinite proof has to recover somehow.

In short, the Phil Reny route solved much of the infinite compactness problem, but the price was a set of conditions that encoded the hard remaining pieces.

9. How the final menu-engine route changed the picture

The later menu-engine route replaced the infinite-dimensional compactness problem with a finite-dimensional payoff-profile problem.

This had two major effects.

First, it removed A5-thick entirely from value optimality. The final theorem gets a value-optimal agent strategy and approximate adversaries under the paper’s standing hypotheses alone.

Second, it replaced A8c-attain by a cleaner **exact-contact** condition. This is still an attainment condition, but it is stated directly in terms of the chosen optimal payoff-profile labeling rather than in terms of a Lusin representative.

For the full robust-rationalizability step, TRE-gen-Hall became **menu-Hall**. This is a weaker and cleaner set-valued calibration condition: instead of forcing one deterministic worst-message map, it allows the adversary to mix over the whole rowwise minimizer correspondence.

The main lesson is that the Phil Reny route was not wasted. It revealed that the real obstruction is not merely compactness of the agent’s strategy space. The hard part is posterior calibration: arranging the adversary’s worst-case message flow so that the induced message-by-message posteriors make the agent’s continuations Bayes-optimal.

10. Takeaway

The Phil Reny route should be understood as a serious but heavier first approach.

It works by:

1. restricting the adviser to regular, τ -dominated kernels;
2. using Balder and Mertens to get a restricted-game maximizer;
3. using Lusin smoothing to lift from restricted adversaries to arbitrary measurable adversaries;
4. adding attainment and Hall-type calibration assumptions to recover full robust rationalizability.

It stalled because the added assumptions were doing real work. The final menu-engine approach keeps the valuable lesson but moves the main existence argument into finite-dimensional payoff-profile geometry, where value optimality becomes unconditional and the remaining calibration gap is stated more transparently.